

ECOSYSTEM SURVIVAL

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Introduction

The Ecosystem Survival software is an interactive grid-based system designed to simulate wildlife management within a controlled environment. Built on an 8 x 8 grid, the application models the behavior and survival conditions of four forest species: Rabbit, Fox, Turtle, and Deer. The system integrates classical algorithmic strategies with structured data management to provide a dynamic and decision-driven simulation, where each action directly impacts ecosystem stability and species survival.

Results

The system demonstrated high computational efficiency during the testing periods. The AVL tree maintained population control operations at a theoretical complexity of $O(\log n)$, processing high population density updates with low resource consumption per period. The JSON communication bridge between C++ and Python operated without bottlenecks, ensuring a smooth synchronous cycle. Furthermore, the linked list correctly recorded the coordinate history of each species without impacting the performance of the main tree's operation

Goal

The objective is to combine algorithm design with efficient data structures to manage the ecosystem. An AVL tree ensures $O(\log n)$ operations for population control, providing accuracy and performance.

Conclusions

The hybrid architecture proved to be a robust and scalable design strategy. The implementation of the AVL tree reduces performance degradation in high population density scenarios, outperforming conventional linear structures. The JSON bridge facilitates code maintenance while ensuring computational speed in each period. Finally, the simultaneous use of balanced trees for the dynamic state and linked lists for the event history constitutes an optimal architectural pattern to efficiently sustain the ecosystem dynamics.

System Architecture

The system will use a hybrid architecture with a C++ engine and a Python interface. C++ manages the AVL tree and linked list, while Python handles algorithms and interaction. Both communicate through JSON files (input.json and state.json).

Bibliography

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